Internship Report

Of

Low-Cost Three-Channel EEG-Based Mental Fatigue Detection Using Machine Learning

Submitted To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE

By

Roll Number of Student: ____________

Name of Student: SANJEEV KUMAR

Batch: 2026-27

CANDIDATE'S DECLARATION

I, Sanjeev Kumar, do hereby declare that the internship report titled "Low-Cost Three-Channel EEG-Based Mental Fatigue Detection Using Machine Learning" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature: ____________

Student Name: SANJEEV KUMAR

Roll No.: ____________

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program at Svaarogyam Medical Devices Pvt. Ltd. I am thankful for their aspiring guidance, invaluably constructive criticism and friendly advice during this work. I am sincerely grateful to Dr. Urvashi Shukla and the entire team at Svaarogyam Medical Devices Pvt. Ltd. for sharing their truthful and illuminating views on a number of issues related to this work, and for providing me the opportunity to work as a Data Science & ML Intern.

I would also like to thank the Department of Computer Science, IILM University, Greater Noida, for their continuous support and for offering the resources and technical guidance required to complete this internship successfully.

I express my gratitude to my parents for their blessings.

Signature: ____________

Student Name: SANJEEV KUMAR

Roll No.: ____________

INTERNSHIP OFFER LETTER / COMMUNICATION PROOF

Svaarogyam
Medical Devices Pvt. Ltd.
Self care made easier

Date: 1/6/2026
To: Sanjeev Kumar

Subject: Internship Offer Letter

Dear Sanjeev Kumar,

We are pleased to offer you an internship opportunity at **our organisation**, based on your academic background and interests. This internship is designed to provide you with hands-on experience and professional exposure in the field of **Data Science and Machine Learning.**

Internship Details:

- **Internship Title**: Data Science & ML Intern
- **Department**: Backend Developer
- **Location**: Remote
- **Duration**: 1/6/2026 to 15/7/2026
- **Working Hours**: 10:00 AM to 5:00 PM, Monday to Friday

Confidentiality & Code of Conduct:

You are expected to maintain the confidentiality of all information and comply with the policies of Company. Any breach of conduct or ethics may lead to termination of the internship.

Certification:

Upon successful completion of the internship and submission of required documentation, a **Certificate of Internship** will be awarded.

We look forward to having you on board and wish you a productive learning experience.

Best regards,

Dr. Urvashi Shukla

Contact Us
+91-9549654239
Jaipur, Rajasthan
svaarogyam@gmail.com

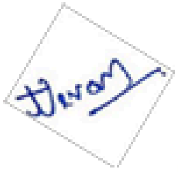
Svaarogyam
Medical Devices Pvt. Ltd.
Self care made easier

CERTIFICATE

of Internship

Is Presented to

Sanjeev Kumar On Successful Completion of his internship in Data Science and Machine Learning with Svaarogyam Medical Devices & Pvt. Ltd.



Best regards,
Dr. Urwashi Shukla
Director & Co-founder

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4. PROJECT DESCRIPTION

4.1 Introduction

Mental fatigue arising from prolonged cognitive effort leads to reduced alertness and impaired decision-making, which can pose serious risks in domains such as healthcare, transportation, and industrial operations. Electroencephalography (EEG) offers a non-invasive means of monitoring brain activity in real time, but commercially available EEG systems are often expensive, bulky, and complex, which limits their use in everyday and wearable settings. During my internship as a Data Science & ML Intern at Svaarogyam Medical Devices Pvt. Ltd., I worked on the design and implementation of a low-cost, three-channel EEG-based system for real-time mental fatigue detection using machine learning. The work combined hardware-level signal acquisition with a complete machine learning pipeline for feature extraction and classification, and was carried out in collaboration with faculty at IILM University, Greater Noida.

4.2 Organization Profile

Svaarogyam Medical Devices Pvt. Ltd. is a healthcare technology company based in Jaipur, Rajasthan, working under the tagline “Self care made easier.” The organisation focuses on developing accessible medical devices and health-monitoring solutions, with an emphasis on affordable, self-care-oriented technology. As a Data Science & ML Intern in the Backend Developer track, I was involved in applying data science and machine learning techniques to physiological signal data as part of the company's ongoing efforts in health-monitoring product development.

4.3 Problem Statement

Unrecognised mental fatigue can lead to accidents, errors, and safety hazards in workplaces that demand sustained attention, such as healthcare, transportation, and manufacturing. Conventional fatigue-assessment methods, such as subjective questionnaires and behavioural observation, are limited in their ability to provide continuous, real-time, and unbiased feedback. While EEG is a reliable physiological measure of fatigue, most existing EEG-based systems rely on high-density, multi-channel hardware and deep learning models that require large datasets and significant computational resources (such as GPUs), making them impractical for low-cost, portable, real-time deployment. There was, therefore, a need for a lightweight EEG-based system that could reliably detect mental fatigue using minimal hardware and a computationally efficient machine learning model.

4.4 Project Objectives

- To design and build an affordable, portable three-channel EEG acquisition system suitable for real-time mental fatigue monitoring.
- To develop a signal processing pipeline that isolates the Alpha (8–13 Hz) and Beta (13–30 Hz) frequency bands relevant to cognitive fatigue.
- To extract meaningful frequency-domain and time-domain statistical features from the EEG signal for use in machine learning models.
- To train and evaluate lightweight machine learning classifiers (Random Forest, XGBoost, LightGBM, SVM) for binary fatigue classification.
- To validate the system using subject-independent testing (Leave-One-Subject-Out Cross-Validation) to ensure real-world generalisability.
- To implement the trained model on a low-power microcontroller (ESP32) for real-time, wearable fatigue monitoring with sub-200 ms inference latency.

4.5 Scope of the Project

The project covers the complete pipeline of a low-cost, wearable mental fatigue detection system: from analog EEG signal acquisition and conditioning, through digital signal processing and feature extraction, to machine learning-based classification and deployment on embedded hardware. The scope was limited to binary fatigue classification (fatigued vs. non-fatigued) using three EEG channels and a dataset collected from 14 healthy participants performing a sustained cognitive task. The system was designed as a proof-of-concept for portable, real-time cognitive-state monitoring in workplace safety, healthcare, and general wellness applications, rather than as a clinical diagnostic tool.

4.6 Technologies and Tools Used

- Hardware: Wet Ag/AgCl electrodes, AD620 instrumentation amplifier, TL072 operational amplifier, ADS1115 16-bit ADC, ESP32 microcontroller, Driven Right Leg (DRL) noise-reduction circuit.
- Signal Processing: 4th-order Butterworth bandpass filtering, Welch's method for Power Spectral Density estimation.
- Programming and ML: Python, scikit-learn (Random Forest, SVM), XGBoost, LightGBM, NumPy/pandas for data handling.
- Validation: Leave-One-Subject-Out Cross-Validation (LOSO-CV), StandardScaler-based feature normalisation.

- Embedded Deployment: ESP32 for wireless data transfer and on-device inference.

4.7 System Architecture

The system architecture consists of three main components. First, the signal acquisition stage records raw EEG data from three channels using wet electrodes, with the AD620 instrumentation amplifier providing high common-mode rejection (CMRR > 90 dB) and the DRL circuit reducing common-mode noise by 20–30 dB. Second, the signal processing and feature extraction stage filters the signal into the Alpha and Beta bands, segments it into 2-second windows with 50% overlap, and computes 24 features per window (8 features across each of the 3 channels), covering band power, the Alpha/Beta ratio, and time-domain statistics such as mean, standard deviation, skewness, kurtosis, and peak-to-peak amplitude. Third, the classification stage feeds these feature vectors into a trained machine learning model, which outputs a binary prediction — fatigued (1) or non-fatigued/alert (0) — along with a probability score, all within a sub-200 ms inference latency on the ESP32.

4.8 Methodology

EEG data were collected from 14 healthy participants (aged 18–22) performing a sustained cognitive task for approximately 60 minutes, while three-channel EEG was recorded at an 83 Hz sampling rate. Fatigue states were labelled using self-reported fatigue levels recorded at regular intervals, resulting in a dataset of 15,189 temporal windows (8,368 non-fatigued and 6,821 fatigued, a 55:45 class split). The raw signal was bandpass filtered and segmented, and 24 features were extracted per window as described above. Four machine learning classifiers — Random Forest, XGBoost, LightGBM, and SVM with an RBF kernel — were trained and evaluated using Leave-One-Subject-Out Cross-Validation, where the model is trained on 13 subjects and tested on the remaining unseen subject, repeated across all 14 subjects to ensure subject-independent evaluation. Feature normalisation was performed within each fold using statistics computed only on the training data, to prevent data leakage. Model performance was assessed using accuracy, precision, recall, F1-score, and area under the ROC curve (AUC), with F1-score used as the primary metric since it balances fatigue detection against false alarms.

Random Forest was found to be the best-performing model, achieving 75.92% accuracy, 84.91% precision, 56.41% recall, an F1-score of 0.6779, and an AUC of 0.7634, outperforming XGBoost (74.73% accuracy, F1 0.6711) and LightGBM (74.97% accuracy, F1 0.6711). The high precision indicates that the system reliably detects genuine fatigue with few false alarms, which is important for building user trust in workplace safety applications, while the moderate recall reflects a conservative classification approach and points to class imbalance and feature-space limitations as areas for future improvement. Compared with a representative deep-learning approach (CNN with continuous wavelet transform, achieving 88.85% accuracy), the proposed three-channel system trades a modest reduction in accuracy for a substantial reduction in hardware complexity, computational requirements (running on an ESP32 microcontroller instead of a GPU), and cost — making it far more suitable for real-time, wearable deployment.

4.9 Expected Outcomes

- A working proof-of-concept three-channel EEG hardware unit capable of acquiring usable EEG signals in the 10–100 µV range.
- A validated machine learning pipeline (Random Forest) achieving 75.92% accuracy and 84.91% precision on subject-independent fatigue classification.
- A real-time inference pipeline on ESP32 hardware with sub-200 ms latency and approximately 45 mW power consumption, suitable for wearable, battery-powered use.
- A low-cost, low-complexity alternative to high-density, GPU-dependent EEG fatigue-detection systems, applicable to workplace safety and general cognitive-state monitoring.

4.10 Certificates of Completion and Other Communication Mail Proof

The internship offer letter issued by Svaarogyam Medical Devices Pvt. Ltd., confirming the internship title, duration, and terms of engagement, is attached in this report (see the Internship Offer Letter / Communication Proof section).

5. BIBLIOGRAPHY / REFERENCES

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